

Problem A. New car

Input file: **standard input**
Output file: **standard output**
Time limit: 4 seconds
Memory limit: 256 megabytes

You are the director of a large monopoly company in 1969, producing ZAG cars. Technology at that time was developing so fast that you decided to urgently release a new ZAG model.

The company has two highly skilled car modellers, Keach and Poch. You instructed them to design a new ZAG model. Keach and Poch decided that the new car model would have to be cross-country due to the specific terrain in the country where you live. As a result, they came to the conclusion that it was necessary to come up with an innovative wheel shape and braking system. After a year of terrain surveys in different regions of the country, Keach and Poch deduced the ideal shape of the required wheel, as well as the relative position of the brake shoe.

The wheel consists of n points (x_i, y_i) located on the coordinate plane and forming a convex polygon. The brake shoe is placed at the point with coordinates (x', y') . **Let's call the center of the wheel the point with coordinates (x_1, y_1) .** When the car moves, the wheel rotates clockwise around the center of the wheel. When the brake is on, the stick moves to the level of the wheel and stops it if at least one edge of the wheel touches it. Keach and Poch are interested in what will be the final position of the wheel when the car comes to a complete stop, if you apply the brakes and go. Or maybe the brakes don't work and the car won't stop.

Input

The first line of the input contains one integer q — the number of test cases.

The first line of each test case contains one integer n — the number of wheel vertices. The vertices are given in the order they were traversed **clockwise**.

The second line of the same test case contains two integers x' and y' — brake coordinates.

The next n lines contain two integers x_i and y_i — coordinates of the i -th vertex of the wheel.

It is guaranteed that the brake is not initially located inside the wheel. In other words, the point (x', y') does not belong to the polygon.

$$1 \leq q \leq 10^5$$

$$3 \leq n \leq 3 \cdot 10^5$$

$$0 \leq |x'|, |y'|, |x_i|, |y_i| \leq 10^9$$

It is guaranteed that the sum of n over all test cases does not exceed $3 \cdot 10^5$.

Output

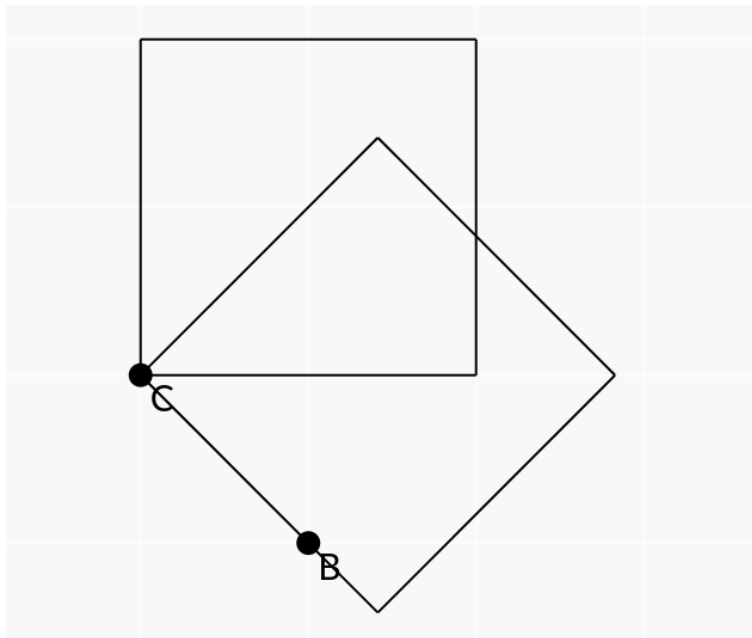
For each of the test cases, print a single integer “-1” (without quotes) in case the car does not stop. Otherwise, you need to output n lines of two numbers each — the final position of the car's wheel when it comes to a complete stop.

Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} .

Example

standard input	standard output
2	0.00000000 0.00000000
4	1.41421356 1.41421356
1 -1	2.82842712 -0.00000000
0 0	1.41421356 -1.41421356
0 2	-2.00000000 0.00000000
2 2	3.18100804 2.67528610
2 0	2.12929146 0.97414171
4	-1.14942781 -0.52585829
0 0	
-2 0	
3 -3	
1 -3	
-2 -1	

Note



Drawing for the first example.

Problem B. Non-sand watches

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Vladimir Yakovlevich bought wrist watch instead of his legendary sand timers. It was hard to get used to hands of the clock, and an interesting problem crossed his mind.

The hands of the clock move continuously (not discrete). Vladimir Yakovlevich wants to know, after how many seconds since the beginning of the day the angle between given pair of hands will be exactly $\frac{p}{q}$ degrees?

Please, help Vladimir Yakovlevich to find all these moments of time from 00:00:00 to 12:00:00

Input

The first line contains two non-negative integers p and q .

The second line contains two integers x and y , encoding hands to find angle between.

1 — second hand, 2 — minute hand, 3 — hour hand.

It is guaranteed that $x < y$.

$$0 < \frac{p}{q} < 360$$
$$1 \leq p, q \leq 10000$$

Output

At first line print one integer n — number of all different answers.

In every of next n lines print two integers k and m such that $\frac{k}{m}$ — irreducible fraction and $\frac{k}{m}$ seconds after start of the day the angle between given hands is $\frac{p}{q}$.

Example

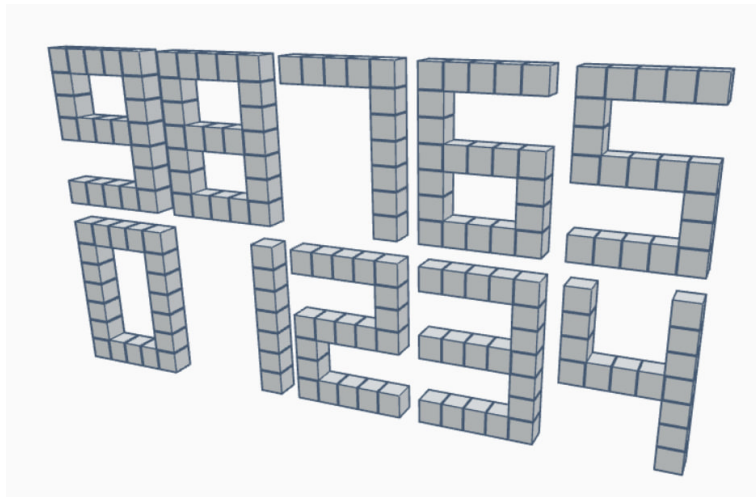
standard input	standard output
1 10	22
2 3	215988 11
	86388 11
	345612 11
	216012 11
	86412 11
	388788 11
	259188 11
	129588 11
	388812 11
	259212 11
	129612 11
	12 11
	431988 11
	302388 11
	15708 1
	432012 11
	43188 11
	27492 1
	172812 11
	43212 11
	475188 11
	345588 11

Problem C. Glass containers

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 256 megabytes

In the country that conquered space, glass container collection points were popular. There were a lot of glasses of different colors, but the most popular were colorless, red R , green G and blue B shades. There was so much glass that it was possible to assemble cubes from them. Exactly 6 glass containers were spent on 6 sides of a cube. The cube could be multi-colored.

Futurism became a reality so quickly that people began to build magnificent machines — integrals. Mathematics was hailed as the most exact science, and over time, people began to celebrate numbers. Every day more and more new, more accurate and larger numbers were invented, and each such discovery was celebrated by the installation of an air monument in honor of this number. The monuments consisted of those same glass cubes and levitated in the air.



(Numbers from cubes of glass containers)

The colored glass started to run low and people decided that cubes that could only show 4 sides would be two-tone. Visible sides from the desired color, and invisible — colorless, of which there is an unlimited resource. However, it is very important that the entire monument externally consists of only one specific color. You are — “C-113” and you have an important mission: you need to find out what flowers can be used to assemble a new monument to the largest and most exact number n !

Since we still have stocks of one-color cubes in different colors, we need to calculate how many two-color cubes with 4 visible sides need to be made.

Input

The first line takes an integer n as input.

The next three lines contain 3 numbers R , G , B — the number of glass containers for red, green and blue colors respectively.

$$0 \leq n < 10^{10^6}$$
$$0 \leq R, G, B \leq 10^9$$

Output

Print “Red” (without quotes) — if red glass container is enough to create cubes from 4 visible sides,

“Green” (without quotes) — if green glass container is enough, “Blue” (without quotes) — if blue glass containers are enough.

If several kinds of colors are enough, then print each color on a separate line in the order: “Red”, “Green”, “Blue”.

If none of the colors are enough, print “No” (without quotes).

Examples

standard input	standard output
2022 284 283 285	Red Blue
42 100 250 30	Green
1 15 16 17	No

Problem D. Differences

Input file: `standard input`
Output file: `standard output`
Time limit: 2 seconds
Memory limit: 512 megabytes

You are given two numbers n and k , as well as an array $a_1, a_2, \dots, a_n$ of n non-negative integers.

You need to choose from an array of k numbers, while preserving their order, so that the sum of the absolute differences of neighboring numbers is maximum.

More formally, you need to choose a set of indices $j_1, j_2, \dots, j_k$ such that the value of $\sum_{j=2}^k |a_{j_j} - a_{j_{j-1}}|$ is the maximum possible.

Input

The first line contains two integers n and k .

The second line contains n non-negative integers a_i .

It is guaranteed that $n \times k \leq 10^8$.

$$2 \leq n \leq 10^5$$

$$2 \leq k \leq n$$

$$0 \leq a_i \leq 10^{12}$$

Output

In a single line print a single number — the answer to the problem.

Examples

standard input	standard output
5 3 1 2 3 4 5	4
3 3 1 4 2	5

Problem E. Buying a car

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

— *Comrade, in order to queue up to buy this car, you need to pass our unique test for the ability to drive a vehicle.*

The demand for cars is growing very fast. There was a long queue due to the fact that the factories are not able to meet the demand. To control the size of the queue, a test was introduced, according to the results of which a citizen gets the opportunity to stand in queue and, after a short wait, buy a car.

The test is that the citizen is given two arrays a and b , each of which consists of n positive integers not exceeding m . He needs to determine the number of ways to replace exactly one number in the array a so that the result of the product of the numbers of the array a is not less than the result of the product of the numbers of the array b . You can only replace it with a positive integer not exceeding m , different from the original number.

Since you need to check a large number of tests, you need to develop a program that, given the numbers n and m , as well as the arrays a and b , will determine the correct answer to the problem.

Input

The first line contains two integers n and m — the size of the arrays and the maximum value in the arrays.

The second line contains n integers a_i ($1 \leq a_i \leq m$) — elements of array a .

The third line contains n integers b_i ($1 \leq b_i \leq m$) — elements of array b .

$$1 \leq n \leq 10\,000$$

$$2 \leq m \leq 10^5$$

Output

In a single line print a single number — the answer to the problem.

Examples

standard input	standard output
3 21 1 1 1 2 2 2	42
2 7 1 5 3 7	3

Problem F. Refrigerator

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 256 megabytes

- *Major, to use the refrigerator “Melania”, enter the password.*
- *What else is the password?*
- *Now I’ll tell you everything...*

You are given two numbers n and q and an array a of n positive integers. To open the refrigerator you need to correctly answer q queries:

1. Add query characterized by numbers l , r and x : add x to all numbers with indices from l to r .
2. Query the sum: given numbers l and r find the value $\sum_{i=1}^{r-l+1} fib_i \cdot a_{l+i-1}$ modulo $10^9 + 7$.

Please note that advanced research by our scientists has shown that the *fib* sequence is optimally specified as follows:

$$fib_i = \begin{cases} 1, & i \leq 2 \\ fib_{i-1} + fib_{i-2}, & i > 2 \end{cases}$$

And we ask you to stick to this method.

Input

The first line contains the numbers n and q — the number of elements in the array a and the number of queries, respectively.

The second line contains n integers — elements of array a .

The following q lines describe queries:

For the first type query: ‘+’ character and three numbers l , r , x ($1 \leq l \leq r \leq n$).

For the second type of query: the ‘?’ character and two numbers l and r ($1 \leq l \leq r \leq n$).

$$1 \leq n \leq 2 \cdot 10^5$$

$$1 \leq q \leq 2 \cdot 10^5$$

$$1 \leq a_i \leq 10^9$$

$$1 \leq x \leq 10^9$$

Output

For each ‘?’ query display one number - answer modulo $10^9 + 7$.

Examples

standard input	standard output
5 6 1 1 1 1 1 ? 1 5 + 1 5 1 ? 1 5 + 1 3 2 + 3 5 3 ? 1 5	12 24 62
5 3 8 7 1 3 4 + 1 4 10 ? 1 5 ? 3 4	116 24

Problem G. Vulnerable delivery service

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 256 megabytes

Most recently, the postal service “Kup” was founded. You have to look for vulnerabilities in the innovative mail delivery system for the 1960s in various parts of the country.

“Kup” founded exactly n mail delivery points. Point with number 1 is the main office of the company. Since the country has an unfavorable relief, “Kup” made exactly m routes to deliver mail from one point to another. In other words, for a pair of points u_i and v_i , between which there is a route, it is possible to deliver mail both from u_i to v_i and from v_i to u_i . It is guaranteed that mail can be delivered from the office to any other point. Completely new points are not as fault-tolerant as their predecessors, so they may accidentally stop working.

Suppose we need to deliver mail from the main office to some point u . Then we call point v **invulnerable** if, if this point fails, it will be possible to deliver mail from the main office to point u . Let's call **order** the set of points $u_1, u_2, \dots, u_k$ ($u_i \neq 1$ for all $1 \leq i \leq k$) to which mail must be delivered from the office. Then the **failure** of the system for a given order will be the number of subsets of **vulnerable** items such that after their **failure** it will not be possible to deliver mail to any of the **order** items. Two subsets are different if exists one deliver point, that belongs to one subset and doesn't belongs to the second.

You will be given q queries that look like this:

+ x — a point with the number x is added to the order. It is guaranteed that it was not included in the order.

– x — the point with the number x is removed from the order. It is guaranteed that it was included in the order.

After each it is necessary to output the **fault tolerance** of the system for the given order modulo $10^9 + 7$.

Input

The first line of the input contains two integers n and m — the number of mail delivery points in the company “Kup” and the number of routes.

The i -th of the next m lines contains 2 integers u_i and v_i — it says that there is a route between two points u_i and v_i .

The $(m + 1)$ -th line contains one integer q — the number of orders.

The next q lines contain one character c and one integer x — description of changing the set of items in the order.

It is guaranteed that each order contains **at least 1 and at most 10** shipping points.

$$\begin{aligned}1 &\leq n \leq 10^5 \\ n - 1 &\leq m \leq 10^5 \\ 1 &\leq u_i, v_i \leq n \\ u_i &\neq v_i \\ 1 &\leq q \leq 10^5 \\ c &\in \{+, -\} \\ 2 &\leq x \leq n\end{aligned}$$

Output

After each request, it is necessary to display the fault tolerance of the system for this order modulo $10^9 + 7$.

Examples

standard input	standard output
4 3 1 2 2 3 2 4 2 + 3 + 4	7 13
7 7 1 2 1 3 2 5 2 4 3 5 5 7 3 6 5 + 7 + 4 - 7 + 2 + 5	7 25 7 6 10

Note

In the first test case, the first order contains exactly one delivery point with the number 3. Vulnerable points are all points except point number 4, because if it fails, it will be possible to deliver mail from the office to point number 3. Accordingly, the fault tolerance for this order will be equal to 7 — any subset of vulnerable points, except for the empty one.

The second order contains two items 3 and 4. For him, all points will be vulnerable, since if at least one point fails, then it will not be possible to deliver mail to any of the points. The fault tolerance of this order is 13 because:

- 1) if one of the points of the order fails, then it will be possible to deliver mail to another.
- 2) if no item fails.

Problem H. That chess player

Input file: standard input
Output file: standard output
Time limit: 3 seconds
Memory limit: 256 megabytes

Once Herman's mind was captured by, **WARNING**, his own **glove**. Of course, Herman immediately regained control of his mind, but something did change...

In a fit of rage, he ate a pawn and began to think that he had become a chess knight. However, he immediately realized that in order to return to his normal state, he needed to count the number of ways to get from one cell of the board to another.

Initially, Herman is in cell $(0, 0)$ of an infinite numbered chessboard. **Note that there are cells with negative coordinates.** In one move, he can move in one of four directions:

- 2 cells to the right and 1 cell down,
- 2 cells to the right and 1 cell up,
- 2 cells up and 1 cell to the right,
- 2 cells up and 1 cell to the left.

Help him count the number of ways to get to the cell (n, m) , that is, to the cell n cells to the right and m cells above the cell $(0, 0)$. Two ways are considered different if there is such a number i that on the move i Herman walked in different directions.

Since the answer can be large, output it modulo $10^9 + 7$.

Input

The single line contains integers n and m — the coordinates of the final cell.

$$0 \leq n, m \leq 20\,000$$

Output

Print a single integer — the answer to the problem.

Examples

standard input	standard output
0 0	1
1 1	2
2 2	6

Problem I. Get evidence

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

During the course of the investigation, detectives Keach and Poch came across a very important witness named John. After a long search, they found him in a tavern.

John agreed to tell everything he saw that day if the detectives would help him with the task.

John has several checkered paper sheets of size n rows per m cells. He really likes the figure of 4 cells, forming the letter “T”. The witness is interested in whether he can cut at least $\lceil \frac{n \cdot m}{4} \rceil$ of such figures from this sheet?

Input

The first line contains a single natural number t — the number of sheets of paper.

Each of the next t lines contains two integers n and m separated by a space — sizes of the next sheet.

$$1 \leq t \leq 10^5$$
$$1 \leq n, m \leq 10^9$$

Output

Print t lines, the i -th of which contains “Yes” (without quotes) if the i -th sheet can be cut into the required number of figures, and “No” (without quotes) in otherwise.

Example

standard input	standard output
3	No
69 42	Yes
88 124	No
23 100	

Problem J. Shifts

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 256 megabytes

Keach and Poch have been in line for their favorite artist's new record for 5 hours. They got bored and Keach offered to play Pochu a game.

Keach has an $n \times m$ matrix consisting of non-negative integers. Keach calls the characteristic of the element $a_{i,j}$ XOR (bitwise exclusive or) of all elements $a_{x,y}$ for which the condition $|x - i| + |y - j| \leq 1$. By a characteristic of a matrix, Keach calls a value equal to the sum of the characteristics of all elements of this matrix.

Keach inserts q numbers one by one at the beginning of the matrix. This means that the number x takes the place of the element $a_{1,1}$, and all the old elements of the matrix are shifted to the right. The last element of each line becomes the first element of the next line. The number of rows may increase after some insertion, and the last row may remain blank at some point in time.

Poch's task — find the original matrix characteristic and the matrix characteristic after each insertion.

Since Keach is very fond of large matrices, and Poch is a very slow thinker, he asked you to help with this task.

Input

The first line contains two positive integers n and m — the number of rows and columns of the matrix, respectively.

In the next n lines you are given the numbers $a_{i,1}, a_{i,2}, \dots, a_{i,m}$ — description of the i th row of the matrix.

The next line contains the number q — the number of requests.

Each of the next q lines contains a single integer x — the next integer that Kitsch inserts into the matrix.

$$2 \leq n, m \leq 10^6$$

$$4 \leq n \cdot m \leq 10^6$$

$$1 \leq q \leq 10^5$$

$$0 \leq x, a_{i,j} \leq 10^9$$

Output

Print $q + 1$ lines containing one integer — the characteristic of the matrix before all insertions and after each insertion.

Examples

standard input	standard output
2 3 1 2 3 4 5 6 2 0 2	24 27 21
7 5 3 7 8 1 8 1 7 4 9 5 2 3 6 0 3 10 7 8 0 1 7 5 8 8 8 8 5 7 4 8 6 8 1 4 5 5 0 7 6 5 9	267 309 305 292 331 341